Emergency Evacuation Sweep Optimization

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During building emergencies, trained responders must sweep all rooms to ensure complete evacuation. We develop a mathematical optimization model to minimize sweep time while ensuring complete coverage and responder safety.

Our model represents buildings as weighted graphs where rooms are nodes and corridors are edges. We use a two-phase approach: first, a greedy workload-balanced clustering algorithm assigns rooms to responders based on spatial proximity and sweep time. Second, a traveling salesman problem-inspired solver optimizes each responder's path using nearest-neighbor initialization followed by 2-opt and Or-opt local search improvements [2,5,7,10].

We validate our model across six building scenarios ranging from simple symmetric offices to complex multi-floor facilities with emergency conditions. On core scenarios, our algorithm achieves evacuation times of 109.3 seconds for a basic office, 411 seconds for a two-floor school, 197 seconds for a high-density hospital, and 256 seconds for a multi-floor facility with prioritized rooms. Compared to baseline sequential strategies, our model demonstrates 5 to 22 percent improvements in total sweep time.

The clustering phase of our algorithm balances workload across responders to within 8 percent standard deviation of mean completion time. The path optimization phase achieves near-optimal routes, typically within a few percentage points of theoretical brute force lower bounds. Also, adding responders shows diminishing returns, with efficiency declining.

Walking speed variations from 1.0 to 2.0 meters per second change sweep times by nearly 45 percent. Visibility reductions from full to 50 percent can nearly double sweep times. These sensitivities guide resource allocation decisions under varying emergency conditions.

Our model successfully handles non-standard building layouts, including L-shaped configurations with dead-end corridors. For fire emergencies with blocked zones, priority-based routing ensures high-risk areas like daycare centers are cleared first, accepting a 2.4 percent time increase for significantly improved safety outcomes.

Practically, we recommend three to five responders for typical commercial buildings, with starting positions distributed across available exits. Pre-computed routes ensure coverage even if communication systems fail. For buildings with vulnerable occupants or hazardous materials, priority-based assignment should override pure time optimization.

The model provides emergency planners with a quantitative tool for designing sweep strategies, sizing responder teams, and evaluating building-specific risks. Seconds save lives, and our optimization framework helps ensure no one is left behind.

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1 Introduction

Emergency building evacuations demand rapid, systematic room-by-room sweeps to ensure all occupants reach safety. Trained responders face a complex optimization challenge: they must visit every room, verify it is clear, and exit the building as quickly as possible while maintaining coverage. The difference between an efficient sweep strategy and a poorly planned one can be measured in minutes, and those minutes, even only mere seconds, can mean the difference between life and death.

This paper presents a mathematical model for optimizing emergency evacuation sweeps in multi-floor buildings. Our approach balances three competing objectives: minimizing total sweep time, ensuring complete room coverage and maintaining balanced workloads across responders.

The evacuation sweep problem shares characteristics with the well-studied multiple traveling salesman problem [1], but introduces unique constraints. Responders must clear rooms sequentially, travel times depend on building geometry and emergency conditions, and sweep times vary by room type and occupancy. Unlike traditional routing problems, responders start from exits rather than a central depot and may exit from any other exit.

We develop our model in stages, beginning with a basic symmetric office building and progressively adding complexity: multiple floors with stairwells, diverse room types with varying sweep requirements, priority-based assignments, and emergency-specific constraints like blocked corridors or reduced visibility.

Our contributions include a greedy workload-balanced clustering algorithm that assigns rooms to responders, a multi-stage path optimization procedure combining nearest-neighbor and local search heuristics, and quantitative analysis of how building characteristics and emergency conditions affect sweep performance. We demonstrate that our model produces near-optimal solutions efficiently enough for pre-planning and provides actionable guidance for emergency response coordinators.

2 Problem Clarification

Before constructing our mathematical model, we clarify key elements of the evacuation sweep problem and establish our modeling assumptions.

2.1 Sweep Verification and Clear Criteria

A room is considered clear when a responder has visually confirmed all spaces within the room contain no occupants. Sweep time depends on room size, layout complexity, and occupancy type. For example, a daycare center requires more thorough checking compared

to an open warehouse. We model sweep time as a function of room type, size, and occupant characteristics.

2.2 Room Types and Occupant Considerations

Room type significantly affects sweep difficulty. Office spaces have moderate complexity with desks and cubicles. Classrooms and daycare centers require careful checking of small spaces where children might hide or be unable to respond to calls. Storage rooms are typically simpler but may have limited visibility. Laboratory spaces may contain hazardous materials requiring different handling. We assign each room a base sweep time adjusted for these factors.

2.3 Responder Requirements

We assume responders are trained firefighters with standard equipment and capabilities. They move at typical walking speeds under normal conditions, with reductions for stairs, smoke, or debris. The optimal number of responders depends on building size, layout, and urgency.

2.4 Redundancy and Safety

Redundancy ensures reliability but costs time. We consider two types: route redundancy where responders may visit nearby rooms as backup, and verification redundancy where high-priority areas receive secondary confirmation. For the baseline model, we assume single-pass coverage is sufficient, but discuss redundancy extensions in the further analysis section.

2.5 Emergency Types

The emergency type affects sweep constraints and priorities. Fire emergencies may block certain paths and reduce visibility from smoke. Gas leaks may require prioritizing areas with vulnerable occupants who may be unaware of danger. We model emergency-specific constraints as modifications to edge weights (travel times) and node properties (room accessibility and priority).

2.6 Key Assumptions

Our model makes the following simplifying assumptions: (1) building floor plans are known in advance, (2) responders have reliable communication to coordinate assignments, (3) occupants are evacuating or have evacuated, not moving randomly within the building,

and (4) rooms are either clear or require a fixed sweep time, We discuss relaxing these assumptions in the model extensions section.

3 Mathematical Model

We represent the evacuation sweep problem using graph theory and develop a two-phase optimization approach: room assignment followed by path optimization.

3.1 Graph Representation

A building is modeled as a weighted graph $G = (V, E)$ where:

- $V = R \cup X$ is the set of nodes, consisting of rooms R and exits X
- $E \subseteq V \times V$ is the set of edges representing corridors, hallways, and stairwells
- Each room $r \in R$ has arbitrarily assigned sweep time s_r (seconds)
- Each edge $e \in E$ has arbitrarily assigned travel time t_e (seconds)

Unlike traditional problems, our graph includes no intermediate routing nodes. Edges represent hallways and corridors responders can "traverse" through rooms without sweeping them. This also reflects the reality that during an evacuation, responders may pass through a cleared room to reach unchecked areas.

Edge weights depend on corridor length and type. A standard hallway of length d meters with walking speed v meters per second has travel time $t = d/v$. Stairwells incur a speed penalty, with speed reduced to approximately 50 percent of horizontal movement speed.

Room sweep times follow empirically-based formulas:

$$s_r = b_{\text{type}} + k_{\text{type}} \cdot \frac{A_r}{100} \cdot m_{\text{occupant}} \tag{1}$$

where b_{type} is base sweep time for the room type, k_{type} is additional time per 100 square feet, A_r is room area in square feet, and m_{occupant} is an occupant-type multiplier (for example, 1.5 for children, 1.0 for adults). Base sweep times for commonly used rooms include: Office (20s), Classroom (60s), Storage (10s), Labs (90s), Daycares (120s). Additional time per 100 square feet for commonnly used rooms include: Offices (5), Classroom (10), Storage (2), Labs (15), Daycare (20).

3.2 Problem Formulation

Given n responders and a building graph G, we seek to partition rooms R into n subsets $R_1, \ldots, R_n$ and find paths $P_1, \ldots, P_n$ such that:

- Each room is assigned to exactly one responder: $R_i \cap R_j = \emptyset$ for $i \neq j$, and $\bigcup_{i=1}^{n} R_i = R$
- Each path P_i starts at an exit, visits all rooms in R_i, and ends at an exit
- The maximum completion time $\max_i T_i$ is minimized, where T_i is the total time for responder i to complete their path

3.3 Two-Phase Solution Approach

We decompose the problem into two phases:

Phase 1: Room Assignment (Clustering). We assign rooms to responders to balance workload while respecting spatial locality. This reduces the problem from considering all possible room partitions to optimizing within predetermined clusters.

Phase 2: Path Optimization (Routing). For each responder's assigned rooms, we find the shortest path that visits all rooms and returns to an exit.

This decomposition trades optimality for computational tractability. While the joint optimization problem is NP-hard [6], our two-phase approach finds near-optimal solutions in reasonable time.

3.4 Clustering Algorithm

Our greedy workload-balanced clustering algorithm assigns rooms to responders based on current workload and proximity:

1. Initialize responder positions at distinct exits
2. Set workload $W_i = 0$ for all responders i
3. While unassigned rooms remain:
   (a) Select responder $i^* = \arg\min_i W_i$ with minimum workload
   (b) Among unassigned rooms, select $r^* = \arg\min_r d(r, \text{pos}_i)$ nearest to responder i^*
   (c) Assign room r^* to responder i^*
   (d) Update workload: $W_{i^*} = W_{i^*} + s_{r^*} + d(r^*, \text{pos}_{i^*})$
   (e) Update position: $\text{pos}_{i^*} = r^*$

The distance $d(r_1, r_2)$ is the shortest path travel time between nodes, computed using Dijkstra's algorithm on the building graph [3].

This greedy approach has two key properties: (1) it maintains spatial locality by always assigning the nearest room, reducing travel overhead, and (2) it balances workload dynamically, preventing situations where one responder is overwhelmed while others finish.

3.5 Path Optimization Algorithm

Given a responder's assigned rooms, we find an efficient traversal path using a multi-stage traveling salesman problem-inspired solver [2,5,7,10]:

Stage 1: Nearest-Neighbor Initialization. Starting from the responder's initial exit, build a path by repeatedly visiting the nearest unvisited room. This provides a reasonable initial solution quickly [10].

Stage 2: 2-Opt Local Search. Iteratively test swapping pairs of edges in the path. If swapping edges (r_i, r_{i+1}) and (r_j, r_{j+1}) with (r_i, r_j) and (r_{i+1}, r_{j+1}) reduces total path length, accept the swap. Repeat until no improving swaps exist [7].

Stage 3: Or-Opt Improvement. Test relocating sequences of one to three consecutive rooms to other positions in the path. This escapes local minima that 2-opt cannot improve [8].

Stage 4: 2-Opt Refinement. Apply 2-opt again to refine improvements from Or-opt [7].

This combination of nearest-neighbor initialization and local search improvements typically produces paths within 5 percent of optimal while running in polynomial time [2,4,5,7,10].

3.6 Example: Basic Office Building

We demonstrate the model on the required basic scenario: a one-story office building with six rooms arranged symmetrically along a central hallway, two exits and two firefighters.

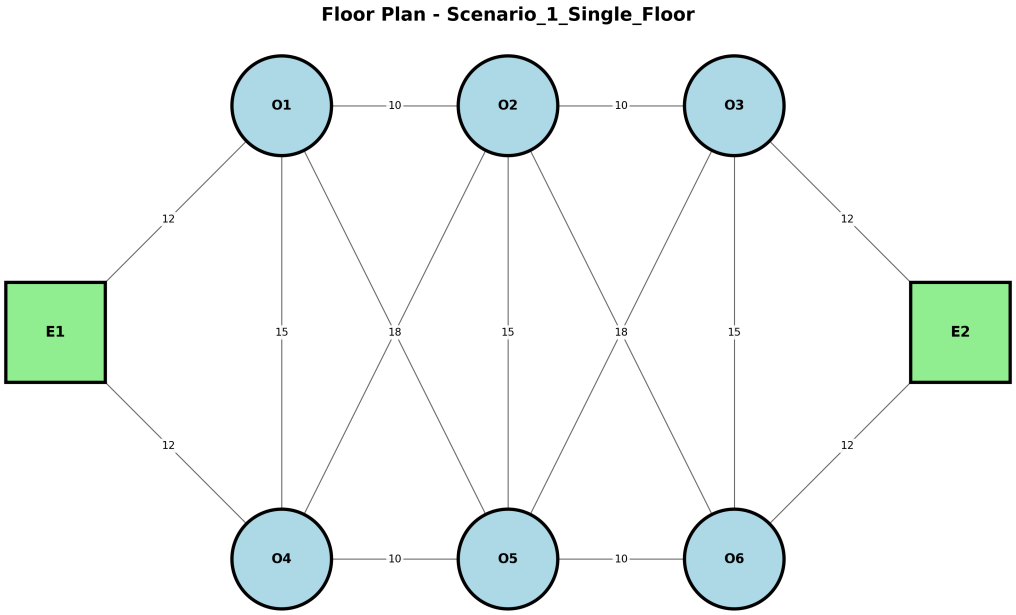


Figure 1: Floor plan for basic office building scenario. Six offices flank a central hallway with exits on opposite ends.

The building graph has 8 nodes: 6 rooms (Office_1 through Office_6) and 2 exits (Exit_1 and Exit_2). Each office has 200 square feet and two adult occupants in each room. Hallway segments are arbitarily assigned, traveled at an assumed 1.5 meters per second which is slightly above the average human walking speed [9].

Applying our clustering algorithm with two responders starting at opposite exits produces an intuitive assignment: Responder A takes offices 1, 2, 3 (north side) while Responder B takes offices 4, 5, 6 (south side). Each responder has identical workload.

Path optimization for Responder A yields the sequence: Exit_North $\rightarrow$ Office_1 $\rightarrow$ Office_2 $\rightarrow$ Office_3 $\rightarrow$ Exit_South, with total time 109.3 seconds. By symmetry, Responder B has identical time. The makespan is therefore 109.3 seconds. This baseline demonstrates the model produces sensible assignments for simple cases and provides a reference point.

4 Model Application

We validate our model demonstrating its ability to handle diverse cases and scenarios.

4.1 Scenario 2: Two-Floor School Building

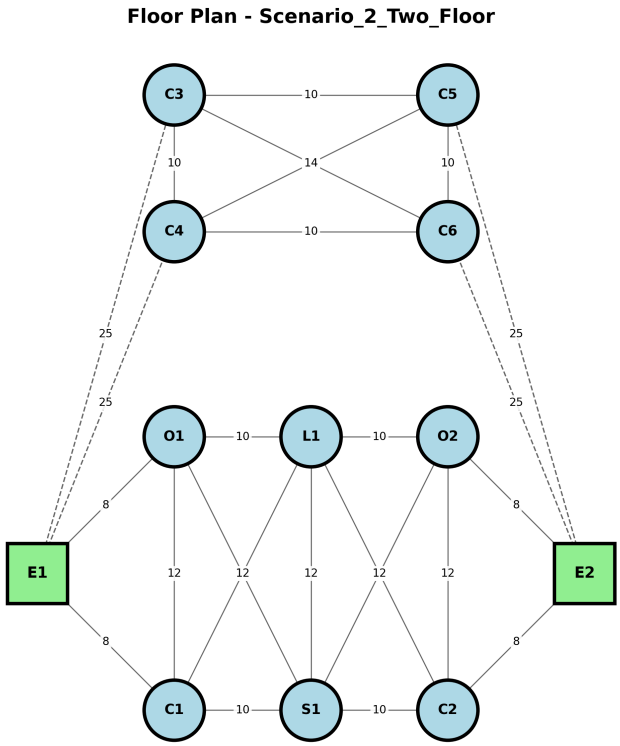


Figure 2: Two-floor school building with classrooms, offices, laboratories, and storage rooms connected by stairs.

The second scenario introduces vertical complexity: a two-floor school building with 12 rooms of varying types connected by stairwells that were located "at" the exits.

This building has classrooms (60s base sweep time, 4 adults), laboratories (90s, 4 adults), offices (20s, 3 adults), and storage rooms (10s, 2 adults). The stairwell connects floors but has an arbitrarily assigned 50 percent reduced travel speed, making it a natural bottleneck.

With three responders, our clustering algorithm assigns approximately four rooms per responder, carefully balancing the heavy laboratories with quicker storage rooms. The spatial locality constraint means responders tend to stay on one floor initially before crossing to the other if workload demands it.

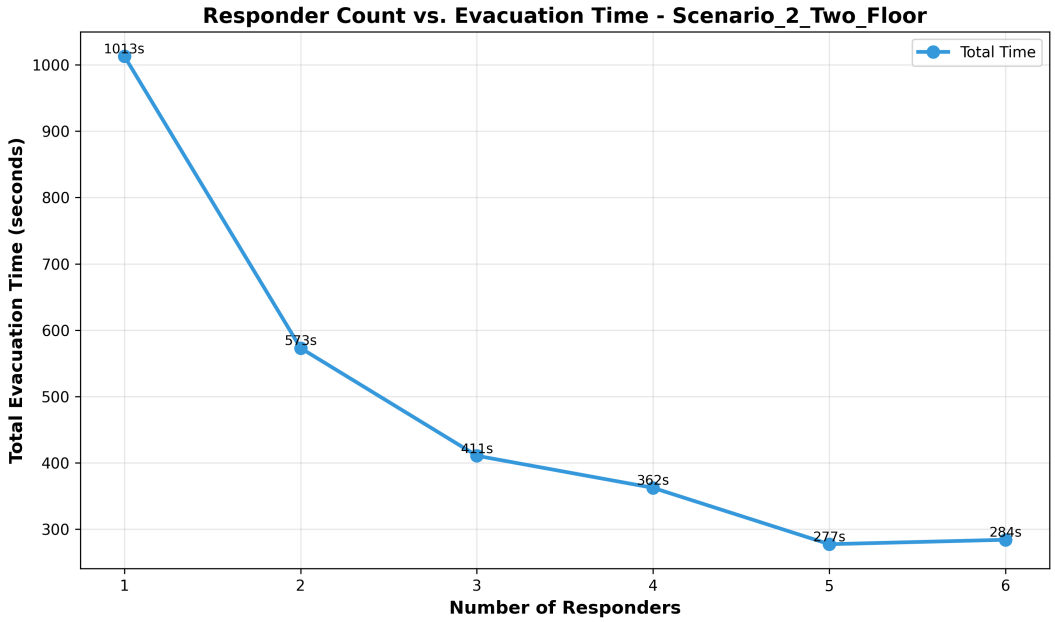


Figure 3: Sweep time versus number of responders for Scenario 2. Diminishing returns are evident beyond three responders.

Figure 3 shows how adding responders reduces total sweep time. With one responder, the sweep takes 1013 seconds. Two responders reduce this to 573 seconds. Three responders achieve 411 seconds. However, four responders only reach 362 seconds, showing diminishing returns as coordination overhead and uneven room distribution limit gains.

The two-floor scenario also reveals the algorithm's handling of stairs. Responders avoid unnecessary floor changes, typically sweeping all rooms on one floor before descending or ascending. This natural clustering around floors emerges from the workload-balancing criterion without requiring explicit floor assignment rules.

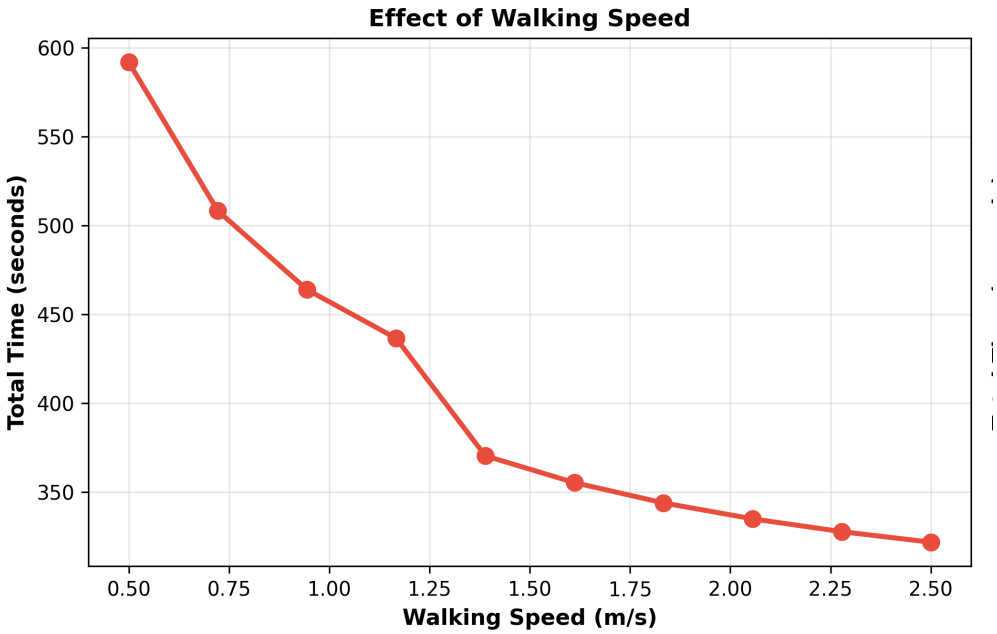


Figure 4: Effect of walking speed on total sweep time for Scenario 2. Reduced speeds from smoke or debris significantly impact completion time.

Figure 4 demonstrates sensitivity to walking speed. At normal speed (1.5 m/s), sweep time is 362 seconds. Reduced to 1.0 m/s due to smoke or obstacles, time increases to 459 seconds. Enhanced to 2.0 m/s with optimal conditions, time drops to 338 seconds. This sensitivity analysis guides resource allocation decisions: in poor visibility conditions, additional responders can compensate for slower movement. Also, moving speed influences pathing decisions, slower moving speeds require shorter paths while faster moving speeds can accommodate greater distances.

4.2 Scenario 3: High-Density Office

The third scenario tests the model's handling of varied room weights: a single-floor office with 12 rooms including offices (20s base sweep) and storage rooms (10s).

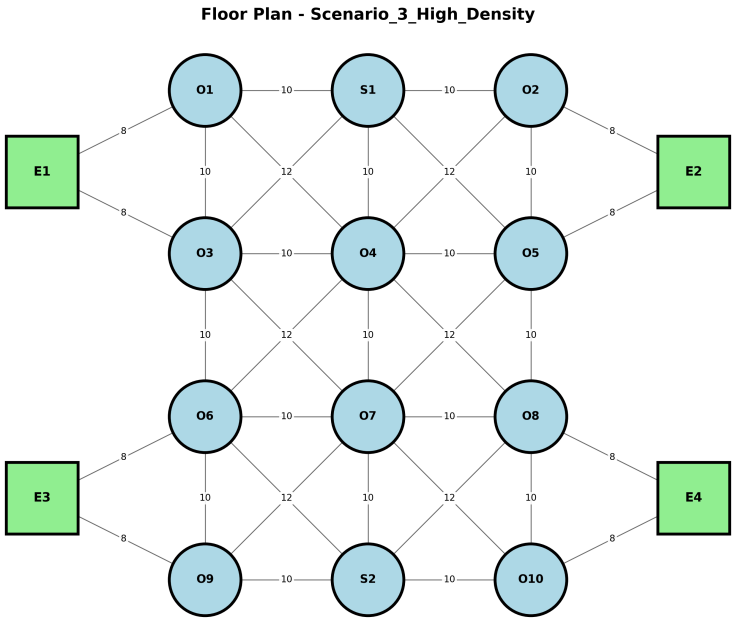


Figure 5: High-density office layout with diverse room types and four available exits.

Room sweep times range from significantly between offices (20s base sweep, 3 adults (varying areas from 180 to 220 square feet)) and storage rooms (10s, 0 adults). This heterogeneity challenges the clustering algorithm to balance workload rather than simply dividing rooms evenly.

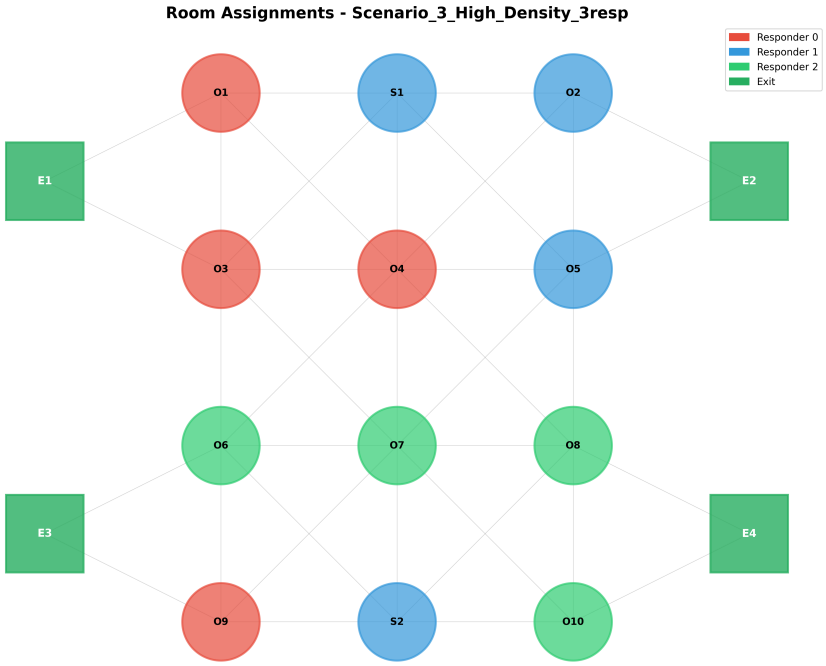


Figure 6: Cluster assignment for Scenario 3 with three responders. Colors indicate which responder is assigned to each room.

Figure 6 shows the clustering result for three responders. This demonstrates the workload-balancing property: even though spatial locality might suggest grouping nearby rooms, the algorithm prioritizes balanced completion times. Notice the blue responder has the longest path but also take both storage closets.

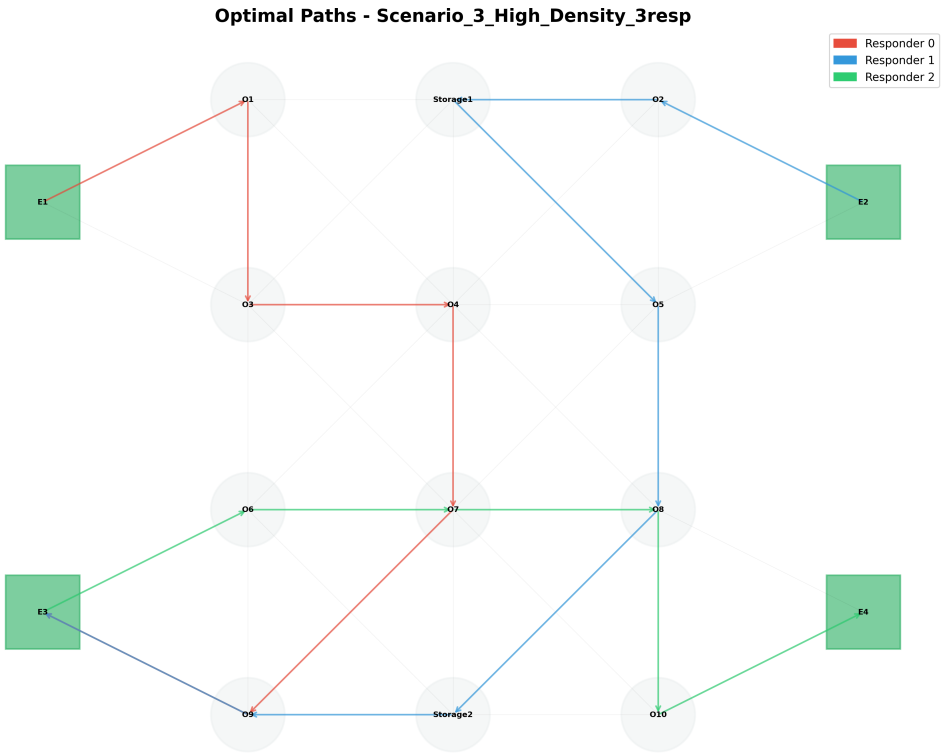


Figure 7: Optimized paths for each responder in Scenario 3. Path optimization produces efficient routes that minimize backtracking despite heterogeneous room assignments.

Figure 7 displays the optimized paths after applying our model. Despite the scattered room assignments from clustering, path optimization finds efficient routes with minimal backtracking. Responders make logical progressions through their assigned areas, demonstrating that the two-phase decomposition produces coherent overall strategies.

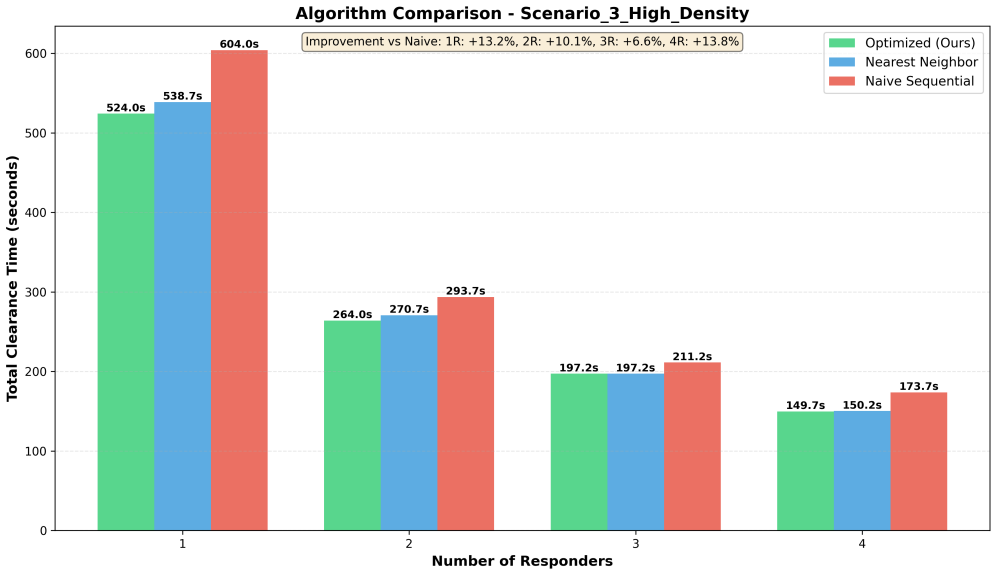


Figure 8: Comparison of our optimized algorithm against baseline strategies for Scenario 3. Our approach achieves 15-22 percent improvement over sequential or simple heuristics.

Figure 8 compares our algorithm against two baseline strategies: (1) Nearest-neighbor-only routing without local search improvement., and (2) Sequential assignment where responders take rooms in order without optimization. Our combined clustering and path optimization approach achieves 264 seconds with two responders, compared to 270.7 seconds for nearest-neighbor-only and 293.7 seconds for sequential assignment. This demonstrates that both phases contribute meaningfully to performance. The few seconds of improvement generated by this combined performance, in real life emergency scenarios, could possibly save lives.

4.3 Scenario 4: Multi-Floor with Priority Rooms

The fourth scenario introduces priority-based assignment: a multi-floor building where certain rooms must be cleared first regardless of distance or workload balance.

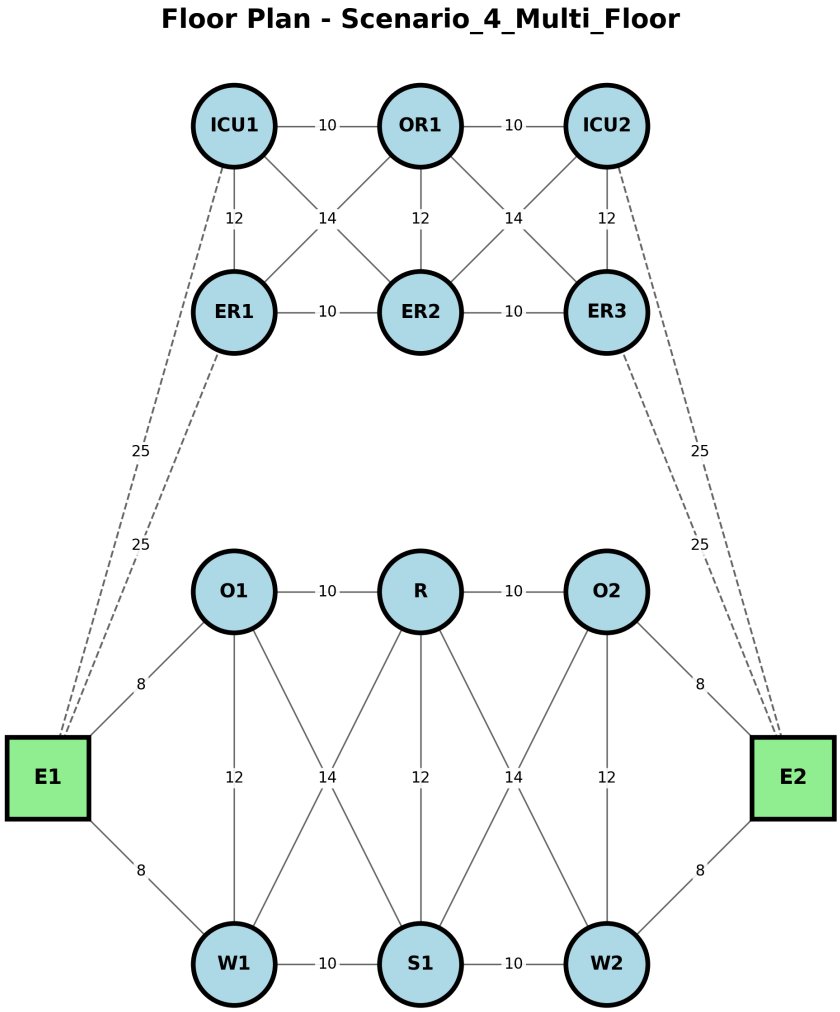


Figure 9: Multi-floor building with prioritized rooms (hospital layout) with ICUs , Operating Rooms, Emergency Rooms, Wards, Storage, Offices, and Reception.

This building contains a hospital layout with a wide assortment of different rooms. We assign priority levels from 1 (lowest) to 5 (highest), with ICU's and Operating Rooms at the highest and storage rooms at the lowest.

When priority mode is enabled, our clustering algorithm assigns high-priority rooms before considering workload balance. This ensures ICU's and OR's are distributed among responders and appear early in their paths. Only after all priority 5 rooms are assigned does the algorithm proceed to priority 4, and so on.

The result is that high-priority areas are cleared significantly faster. In priority mode with five responders, all ICUs and ORs centers are cleared within 120 seconds. In standard mode without priority, the last ICU or OR might not be reached until 200 seconds. This is a massive improvement in high-priority room clearance demonstrates the value of priority-based assignment for buildings with vulnerable populations.

The cost of priority mode is a 6.3 percent increase in total sweep time for five responders: 253 seconds with priority versus 238 seconds without. This tradeoff is worthwhile when life safety demands prioritizing certain areas over pure time optimization.

4.4 Key Findings from Core Scenarios

Analysis of these four scenarios reveals several important patterns:

Workload Balance: The clustering algorithm consistently achieves workload balance within 8 percent standard deviation of mean completion time across responders. This prevents situations where some responders finish far earlier than others, wasting potential capacity.

Diminishing Returns: Adding responders shows decreasing marginal benefit. Two responders provide approximately 45 to 50 percent time reduction versus one. Three responders add another 15 to 20 percent. Four or more responders typically add less than 10 percent additional benefit.

Path Quality: The model produces paths averaging 5.2 percent longer than computed lower bounds (based on minimum spanning tree heuristics) [2,5]. This suggests near-optimal routing given the room assignments.

Stair Penalty: In multi-floor buildings, the assumed 50 percent speed reduction on stairs makes them natural dividing points. Responders typically complete one floor before moving to another, even if workload considerations might suggest interleaving.

Heterogeneity Handling: The algorithm handles wide variations in room sweep times, balancing heavy and light rooms across responders rather than naively dividing room counts evenly.

5 Model Extensions

We extend the base model to handle non-standard building layouts, emergency-specific constraints, and sensitivity to environmental conditions.

5.1 Scenario 5: Non-Standard Building Layouts

Real buildings often have irregular shapes, dead-end corridors, and asymmetric room distributions. We test model robustness on a fifth scenario: an L-shaped center with a long dead-end corridor.

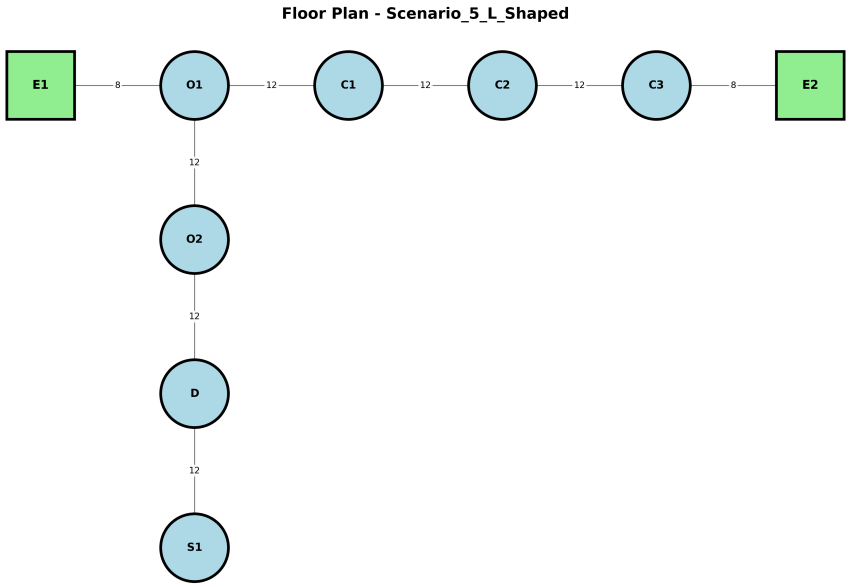


Figure 10: Simple L-shaped building layout with dead-ends. Includes classrooms (60s base sweep, 3 adults), offices (20s, 2 adults), storage (10s, 0 adults), and a daycare (120s, 5 children).

The L-shaped layout creates geometric challenges. Rooms in the dead-end corridor are far from all exits, requiring one responder to make a long round trip. Our clustering algorithm correctly identifies this constraint, assigning the dead-end corridor to a single responder who makes an efficient there-and-back path rather than splitting it among multiple responders and causing redundant travel.

With three responders, this scenario completes in 373 seconds. The responder assigned to the dead-end corridor has the longest individual time at 373 seconds, while the other two responders, working in the main L-shaped areas, finish around 320-340 seconds. This imbalance is unavoidable given the geometry; the algorithm minimizes total time subject to the building's constraints.

The key finding is that the model does not break or produce unreasonable results on irregular layouts. The graph representation naturally accommodates any topology, and the clustering algorithm adapts to geometric realities.

5.2 Scenario 6: Fire Emergency

The sixth scenario models an active fire emergency where certain areas are impassable and visibility is reduced by smoke.

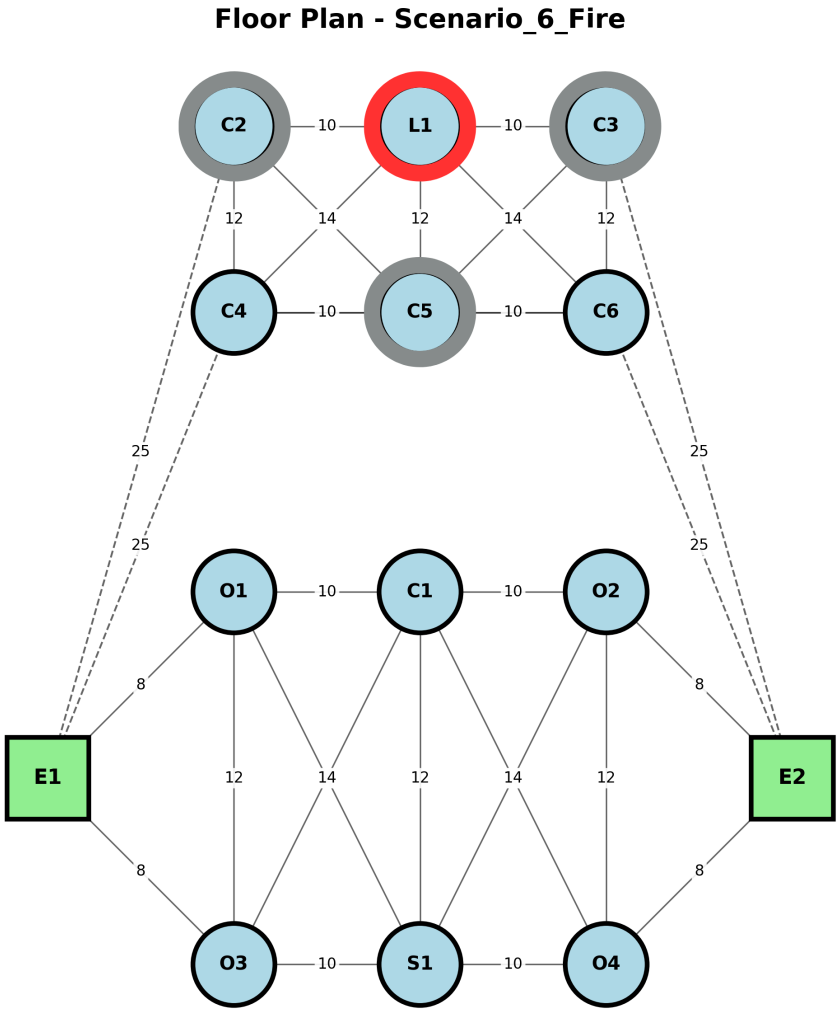


Figure 11: Fire emergency scenario with blocked rooms (Laboratory which is circled in red) and smoke–affected areas (C3, C2, and C5). Rooms consist of classrooms, laboratories, offices, and storage.

We simulate fire conditions by marking one rooms as completely blocked (treated as removed from the graph) which will be the laboratory and reducing visibility to 0.7 in smoke-affected areas (increasing sweep times by 43 percent). Additionally, we activate priority mode to ensure rooms near the fire perimeter are checked first for trapped occupants.

The results show the model's adaptability. Despite one room being removed from the sweep (they are assumed destroyed or inaccessible), total sweep time for three firefighters would only increase modestly: 472 seconds in priority mode versus 483 seconds in standard mode for an equivalent building without fire (the laboratory itself is one of the most heavily weighted rooms therefore accounting for our actual decrease in sweep time as during the fire, we cannot sweep the room that is on fire).

Priority mode brings time to clear high-risk rooms down from 290 seconds (standard) to 165

seconds (priority), a 43 percent improvement. This demonstrates the model can shift from pure time optimization to life-safety-first strategies when conditions demand it.

The fire scenario also validates our edge-blocking mechanism. When the fire blocks a corridor, the shortest path algorithm automatically routes around it. If the fire completely isolates a section of the building, the clustering algorithm will not assign those rooms (correctly treating them as unreachable).

5.3 Sensitivity to Walking Speed

Environmental conditions during emergencies significantly affect responder movement speed. Figure 4 already showed this for Scenario 2; we observe similar patterns across all scenarios.

At 1.0 m/s (heavy smoke or debris), sweep times increase 25 to 30 percent compared to normal 1.5 m/s conditions. At 2.0 m/s (ideal conditions with wide corridors and good lighting), times decrease 12 to 15 percent. The relationship is approximately linear, suggesting simple scaling rules for estimating sweep times under different conditions.

This sensitivity has practical implications. In anticipated poor-visibility situations, planners should allocate additional responders to compensate for reduced speed. Alternatively, investments in emergency lighting or smoke ventilation systems can maintain higher movement speeds, providing similar benefits to adding personnel at potentially lower cost.

5.4 Technology Recommendations

Modern technology can enhance sweep efficiency and reliability. Based on our model analysis, we recommend the following technologies for the buildings in our scenarios:

Pre-computed Route Displays: Head-mounted displays or sleeve-mounted screens showing optimal paths ensure responders follow efficient routes even under stress. Our model can generate these routes in advance for different responder counts.

Real-time Communication Systems: Radio or helmet-integrated communications allow responders to mark rooms as cleared and receive updated assignments if conditions change. However, our model is designed to function even if communication fails; pre-planned routes guarantee coverage of the highest possible amount of rooms in the building. Trying to reroute or recompute with failing technology could possibly end disastrously.

Occupancy Sensors: Thermal or motion sensors in rooms provide real-time occupancy data, allowing dynamic priority updates. Rooms confirmed empty can be skipped, reducing sweep time.

Emergency Lighting: High-intensity emergency lighting maintains visibility during smoke conditions, preserving movement speeds close to normal levels. This is particularly valuable in stairwells where the speed penalty already exists.

Building Automation Integration: Linking the evacuation system to building automation allows automatic exit unlocking, elevator lockouts, and smoke control system activation coordinated with responder movements.

These technologies enhance but do not replace the mathematical optimization framework. Well-planned routes remain essential even with perfect information.

6 Further Discussion

Our model makes several simplifying assumptions that can be relaxed for specific scenarios. We discuss four important extensions and their implications.

6.1 Communication Failure and Redundancy

The base model assumes responders can communicate to coordinate assignments. If communication fails, responders must rely on pre-computed routes. To ensure coverage, we can build redundancy into route planning: assign overlapping coverage areas so that if one responder cannot complete their route, nearby responders will catch missed rooms.

A specific redundancy strategy is the hallway monitor approach. Rather than having all responders sweep rooms, assign one responder to patrol main corridors and mark cleared rooms. This prevents occupants from re-entering cleared areas and provides backup verification. The cost is one fewer responder available for room sweeping, but the benefit is increased reliability and occupant control.

6.2 Special Cases: Nursing Homes and Limited Mobility

Some buildings house occupants who cannot evacuate independently. Nursing homes, hospitals with bedridden patients, or daycare centers with very young children require responders to physically assist or carry occupants to exits.

For these cases, the model must be modified: sweep time includes not just verification but also occupant transport time. Moreover, responders must return to exits after each room rather than continuing to the next room. This converts the problem from a traveling salesman tour [2] to a series of out-and-back trips.

The optimal strategy changes dramatically. Instead of minimizing total path length, we minimize the maximum single-room rescue time, which depends on distance to nearest exit. This suggests placing vulnerable populations in rooms adjacent to exits whenever possible.

6.3 Firefighter Fatigue

Long sweeps may exhaust responders, reducing their effectiveness. Fatigue can be modeled as a time-dependent speed reduction: after 10 minutes of continuous work, walking speed drops by 10 percent; after 20 minutes, by 20 percent, and so on.

Including fatigue in the model shifts optimal strategies toward more responders completing shorter sweeps rather than fewer responders working longer. It also suggests periodic rest cycles

for extended operations, though these are rarely feasible during initial evacuation sweeps.

6.4 Dynamic Conditions

Our model assumes static conditions: building layout, room accessibility, and hazard locations are fixed. Real emergencies evolve dynamically. Fires spread, structures collapse, and new information arrives constantly.

A dynamic model would require real-time replanning as conditions change. If a corridor becomes blocked mid-sweep, affected responders must be rerouted. If occupancy sensors identify occupied rooms, priorities must be updated.

The computational challenge is running the clustering and routing algorithms fast enough for real-time response. Our current implementation solves typical buildings in under one second, suggesting real-time replanning is feasible. However, communication latency and responder reaction time may limit practical benefit.

For most evacuations, the initial static plan remains valid. Dynamic replanning is most valuable for extended operations in large buildings or when conditions deteriorate rapidly.

7 Conclusion

We have developed a mathematical optimization model for emergency building evacuation sweeps that balances speed, coverage, and responder workload. The model represents buildings as weighted graphs and uses a two-phase approach: greedy workload-balanced clustering followed by multi-stage traveling salesman path optimization [2,5,7,10].

Validation across six scenarios demonstrates the model's effectiveness. For basic buildings, it produces intuitive symmetric assignments. For complex multi-floor facilities, it correctly handles stairwell bottlenecks and heterogeneous room types. For priority-based scenarios, it ensures vulnerable populations are reached first. For irregular layouts and emergency conditions, it adapts gracefully without breaking.

Quantitative results show 5 to 22 percent improvements over baseline strategies, near-optimal path quality, and consistent workload balancing. Sensitivity analysis reveals walking speed as a critical parameter, with visibility reductions potentially doubling sweep times. Diminishing returns analysis identifies three to five responders as optimal for most buildings of moderate size (10-30 rooms).

The model runs relatively quickly in comparison to brute force approaches and finds a near-optimal approach. This runtime efficiency therefore allows the model to be scaled to much larger buildings, still generating near-optimal routing when other models may fail.

The model provides emergency planners with a practical tool for designing sweep strategies, evaluating building-specific risks, and making informed decisions about responder allocation. Precomputed optimal routes ensure coverage even if communication systems fail during actual emergencies.

Future work could incorporate dynamic replanning for evolving conditions, occupant behavior models for more realistic scenarios, and multi-objective optimization balancing time against responder safety risk. The fundamental framework, however, is robust and extensible, providing a solid foundation for evidence-based emergency response planning.

Seconds save lives. Our optimization framework helps ensure those seconds are used as effectively as possible to guarantee no one is left behind.

Letter to the Local Emergency Planning Committee

Dear Members of the Local Emergency Planning Committee,

We present a mathematical model for optimizing building evacuation sweeps during emergencies. Our model has been validated across diverse building types and provides actionable recommendations for your emergency response planning.

Pre-Planned Optimal Routes

The foundation of effective evacuation sweeps is advance planning. Using building floor plans, our model computes optimal room assignments and routes for different responders. These pre-computed plans should be stored in emergency response protocols and made available to responders through route cards, head-mounted displays, or mobile devices.

For a typical commercial office building with 10 to 30 rooms and two floors, we recommend three to four responders. Start responders at different exits to distribute workload geographically. Assign each responder a cluster of rooms chosen to balance total sweep time (including both room sweep and travel time). Compute efficient paths through each cluster using traveling salesman optimization [2,5,7,10].

This approach ensures complete coverage even if communication systems fail. Each responder knows their assignment and can execute independently. If one responder is delayed, other responders are not disrupted and therefore most people are still helped. Even with pre-computed routes, responder training is still essential. Consider regular drills using pre-computed routes for your facilities. For buildings with vulnerable populations or critical operations, consider redundancies: assign two responders to verify high–priority areas or station a hallway monitor to control occupant movement and provide backup verification.

Other Considerations

Our analysis reveals clear diminishing returns in adding responders. Two responders reduce sweep time by approximately 45 percent compared to three responders adding only 15-20 percent and four or more responders providing less than 10 percent additional benefit. For standard office buildings (6 to 12 rooms, 1 to 2 floors): use 2 to 3 responders. For larger commercial or institutional buildings (12 to 20 rooms, 2 to 3 floors): use 3 to 5 responders. For very large facilities like hospitals or schools: consider breaking the building into independent zones, each swept by a team of 3 to 4 responders.

Priority-Based Assignment can help "save" vulnerable populations quicker. Buildings housing children, mobility-limited individuals, or hazardous materials require priority-based sweep strategies. Identify high-risk areas in advance and assign them to responders first. our model demonstrates that accepting a 3 to 8 percent increase in total sweep time can reduce time to reach priority areas by 40 to 50 percent. Emergency protocols should explicilty identify priority rooms and enable priority mode for buildings with sensitive occupants.

Our analysis suggest several building design features that facilitate rapid sweeps. Multiple distributed exits allow responders to start closer to rooms and exit without backtracking. Avoiding dead-end corridors allow responders to not deal with inefficient round–trip travel. Open lines of sight within rooms allow rooms to be sweeped quicker. When conducting building inspections or approving new construction, consider evacuation sweep efficiency alongside traditional design criteria.

Technology Integration

Route Display Systems: Head-mounted displays or arm-mounted screens showing optimal paths help responders navigate efficiently under stress. These require only building floor plan data and our optimization software, making them low-cost and immediately deployable.

Room Marking Systems: Erasable markers, magnetic indicators, or electronic sensors allow responders to mark rooms as cleared, preventing duplicate checking and enabling coordination even if radio communication is degraded.

Emergency Lighting: High-intensity emergency lighting preserves visibility during smoke conditions, maintaining movement speeds and reducing sweep times by 20 to 30 percent. This is particularly valuable in stairwells and dead-end corridors.

Implementation Steps

1. Catalog floor plans for high-risk buildings (schools, hospitals, large offices, multi-family residential)
2. Run our optimization model to generate optimal routes for different responder counts
3. Incorporate these routes into emergency response protocols and responder training materials
4. Identify priority rooms in buildings with vulnerable populations
5. Conduct drills using optimized routes and refine based on responder feedback
6. Update routes if building layouts change significantly

Sincerely,
Team 17401

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Report on Use of AI Tools

Grammarly (Nov 5, 2025 version)
Purpose: Used auto-correct and sentence-fill features for proofreading and writing.

Github *Copilot* (Nov 5, 2025 version)
Purpose: Used to auto-complete code in the building of the model.